OCCPL Innovations

Multi-Product SaaS Platform

Phase plan and project status report

Overall	All 10 phases and Billing (Razorpay) complete in code. The AWS hosting is fully defined and rehearsed; it has not yet been switched on in OCCPL's AWS account.
Progress	10 of 10 phases plus billing complete in code; going live needs the AWS account set-up (see Next step).
Tests passing	Backend: 80 unit/slice tests and 77 integration tests against real MySQL 8.4 (the production version; also run on 9.7), Keycloak 26.8.1, Redis 8.10 and SMTP; 2 sample-product tests; 1 end-to-end test across Keycloak, Core and the sample product. Frontend: 49 tests, lint, format and production build; browser walkthrough of both portals against the Docker stack with real Keycloak sign-in and MFA. Load test with 1 million users and a backup/restore drill with measured results. Production rehearsal: the AWS start-up scripts brought the full system up on MySQL 8.4 behind a stand-in load balancer, with real sign-in and MFA through the public addresses.
Try it	Local Docker stack: API docs with Keycloak sign-in at http://localhost:8080/swagger-ui.html; every outgoing email (e.g. invitations) in the Mailpit inbox at http://localhost:8025; sample Notes product API at http://localhost:8090. Admin portal (Super Admin and Company Admin) at http://localhost:4200.
Not yet verified	Nothing has been created in AWS yet (no account access from the development sandbox): the Terraform code is validated, not applied. Encrypted Redis was not part of the rehearsal. Razorpay itself is not reachable from the sandbox: billing was tested against a stand-in that follows Razorpay's published API, so a test-mode payment must be tried before going live.
Next step	Set up AWS: account access, the occplsoftech.com DNS zone in Route 53, email sending approval (SES), GitHub settings; then the production environment, tested with a rollback and a database restore before launch (infra/aws/RUNBOOK.md). Before live payments: decide GST handling (D-11) and run one Razorpay test-mode payment. Before sizing production: confirm the service targets (D-6) and repeat the load test on the production environment before launch.

Phase overview

#	Phase	Status	Scope
1	**Discovery & architecture**	Done	Architecture, data model, security model, API inventory, backlog, ADRs
2	**Foundation**	Done	Backend & frontend skeletons, Docker stack, CI, error handling, observability
3	**Identity & access**	Done	Keycloak login mapping, roles & permissions, company-level checks, MFA, rate limiting
4	**Company & membership management**	Done	Company lifecycle, invitations, member roles, audit trail
5	**Product catalogue**	Done	Product registration, versioned configuration, features
6	**Plans, subscriptions & entitlements**	Done	Plans, prices, limits, subscription lifecycle, usage enforcement
7	**Provisioning & sample product**	Done	Reliable provisioning with retries; reference product service

#	Phase	Status	Scope
8	**Frontend portals**	Done	Super Admin and Company Admin screens
–	**Billing (Razorpay)**	Done	Gateway invoices, payment webhooks, payment-driven renewal
9	**Hardening**	Done	Security review, isolation suite, load tests, backup drill
10	**Deployment**	Done	AWS infrastructure as code, release pipeline, monitoring, backups, runbook

Decisions and technology

Topic	Decision
Versions	Latest stable; latest LTS where a choice exists (ADR-0007)
Login provider	Self-hosted Keycloak (identity data stays in OCCPL infrastructure); MFA required for OCCPL staff
Payments	Razorpay; invoices generated by Razorpay, Core stores references
Grace period / retention	7-day grace before suspension; data kept 90 days after cancellation/expiry (configurable)
Architecture	Modular monolith for the Core Platform; products are independent services
Domain	occplsoftech.com: admin., api., auth. and one address per product
Staging	No staging environment on AWS: releases are tried on the local Docker setup, then approved for production
Hosting	AWS Mumbai (ap-south-1): servers running Docker behind a load balancer with firewall (WAF), managed MySQL (RDS), managed Redis, AWS email (SES)

Component	Version	Note
Java	25 LTS	Java 27 is newer but has ~6 months of support
Spring Boot	4.1.1	Spring Framework 7, Security 7, Hibernate 7.4, Jackson 3
Angular	22.2	TypeScript 6, Vitest, zoneless
MySQL	8.4 LTS	Amazon RDS in production; tests also run on 9.7 for the later upgrade
Keycloak	26.8.1	Self-hosted
Redis	8.10	Rate limiting (Phase 3), caching later
Nginx	1.30	Unprivileged container for the portal

Phase details

Phase 1 — Discovery & architecture

Done

Delivered / scope

- Project scope, assumptions, open decisions and risks (PROJECT.md)
- Module boundaries and dependency rules, integration contract for product services (ARCHITECTURE.md)
- About 35-table data model, subscription state machine, usage-limit rules, migration strategy (DATABASE.md)
- About 80 API endpoints with required permissions; standard error format (API_SPEC.md)
- Authentication, authorization and tenant-isolation design (SECURITY.md); 7 architecture decision records

Phase 2 — Foundation

Done

Delivered / scope

- Backend skeleton: standard JSON errors with tracking IDs, login-token validation, CORS allowlist, security headers
- Health checks and metrics on an internal port; structured JSON logs; database migrations
- Portal shell: runtime configuration, API error handling, design tokens (light/dark), lint and formatting
- Docker Compose stack (MySQL, Keycloak, backend, portal); GitHub Actions CI; weekly dependency updates

Verified

- 22 unit/slice tests and 5 integration tests (MySQL 9.7); 14 frontend tests; production build
- Full stack started locally: real Keycloak tokens accepted; wrong audience, issuer or tampered tokens rejected
- Portal loaded in a browser at phone width with no errors (a CSP issue was found and fixed)

Phase 3 — Identity & access

Done

Delivered / scope

- Database tables for users, identities, profiles, companies, memberships, roles, permissions and audit log
- Seeded catalogue: 28 permissions and 5 built-in roles (Super Admin, Platform Support, Company Admin, Manager, User)
- Sign-in mapping: users created or linked on first login only with a verified email; disabled accounts blocked; an email cannot be taken over by a second Keycloak account
- First Super Admin set by a one-time configured email; every change audit-logged
- Two-factor login for OCCPL staff: platform permissions withheld until the login used an authenticator app
- Platform role management API; the last Super Admin cannot be removed
- Company-level checks: another company's data returns 'not found'; suspended companies are read-only
- Endpoints: GET /api/v1/me, PUT /api/v1/me/profile, GET /api/v1/permissions
- Keycloak admin client (create users, required actions, enable/disable, logout)
- Rate limiting per user / per IP using Redis; requests still allowed if Redis is down (failure is counted)

Verified

- 6 application integration tests incl. migrations and seed data on MySQL 9.7

- 15 tenant-isolation tests: cross-company access, role limits, suspended/deactivated states, MFA rule, first-login rules, concurrent first login, last-Super-Admin rule, profile validation
- 3 Keycloak tests: admin client via service account; token audience/issuer checks; first Super Admin with authenticator-app enrolment and OTP login through Keycloak's real login pages
- 2 rate-limit tests with real Redis plus a fail-open test
- Docker stack: sign-in from Swagger UI through Keycloak in a real browser, GET /api/v1/me

Open item
- A dedicated expired-token test (expiry is enforced by the framework's default validator)

Phase 4 — Company & membership management

Done

Delivered / scope
- Companies: create with primary contact and initial Company Admin invitation in one step; search, edit (company admins may change only the display name), activate/suspend/deactivate with history
- Company settings incl. allowed email domains for invitations; contacts (primary, billing, technical)
- Invitations: invite, resend, revoke, accept, automatic expiry; links are one-time and stored only as hashes
- New users get a Keycloak 'set your password' email that returns to the accept link; existing users get the platform's invitation email
- Members: list/search, suspend/reactivate/deactivate, change roles — nobody can grant more than they have, change their own membership, or remove the last Company Admin
- Platform users: list, disable/enable (also in Keycloak), memberships across companies
- Audit log search for OCCPL staff and per company for company admins
- Email outbox with retries; local Mailpit inbox shows every email

Verified
- 13 company/membership tests incl. every negative rule; 3 email outbox tests (delivery, retry, rollback)
- 2 end-to-end invitation tests with real Keycloak and SMTP: new admin sets password via the emailed link and joins; existing user accepts the platform email
- Docker stack: company created over the API, invitation email visible in Mailpit

Open item
- Custom company roles; branded Keycloak email text; Idempotency-Key header; seat limits (Phase 6)

Phase 5 — Product catalogue

Done

Delivered / scope
- Register products (start as draft) with app URL, API URL and the identifier their service accepts in tokens
- Edit, activate (only after a configuration is published), retire and reinstate products; every change audited
- Versioned configuration: each version is checked against a published schema, can never be edited, and can be published or rolled back; configurations that contain secrets are refused
- Features per product: on/off features and limits (seats, API calls, storage, ...) with reset periods

Verified
- 5 configuration-validation tests and 3 URL-rule tests
- 4 catalogue integration tests: registration rules, edits, version publish/rollback, feature rules

Open item
- Per-company values for product settings (Phase 7); product logos (needs safe file uploads)

Phase 6 — Plans, subscriptions & entitlements

Done

Delivered / scope

- Plans with prices per currency, included products and feature limits; published plans can't change (copy instead); archive
- Assign plans to companies with an optional trial; upgrade immediately, downgrade at the end of the period; renew, cancel now or at period end, resume; full history and audit
- Automatic lifecycle every 5 minutes: trial ends, unpaid periods become past due, 7-day grace then suspension, 90 days suspended then expiry; data kept 90 days after cancellation or expiry
- Entitlements follow subscriptions instantly; company suspension blocks access; usage limits counted safely even under heavy parallel use; seat limits stop invitations beyond the plan

Verified

- 38 state-machine tests: every status/event combination
- 10 integration tests incl. plan rules, lifecycle sweeps, plan changes, entitlement merging, usage replay, and 120 parallel requests against a limit of 50 (exactly 50 allowed)

Open item

- Per-product seats (product-service endpoints were delivered in Phase 7, payment-driven renewal with billing)

Phase 7 — Provisioning & sample product

Done

Delivered / scope

- Product services sign in to Core with their own Keycloak client; Core limits each one to its own product
- Service API for products: is this user of this company allowed to use the product or feature? Report usage against limits (safe to repeat)
- When a company gets a product, Core sets the company up in the product automatically; failed calls are retried with increasing waits, permanent failures alert the operations team and can be retried by an admin
- When access ends, the company's data in the product is removed only after the 90-day retention period, and kept if the company subscribes again before then
- Company product list with launch links once the product is ready
- Sample "Notes" product with its own database, showing the full integration

Verified

- 6 integration tests with a fake product: success, slow product, server errors, rejected requests, asynchronous completion, wrong-product callbacks, delayed removal and its cancellation
- 2 sample-product tests; 1 end-to-end test with real Keycloak, Core and the sample product as a separate program with its own database

Open item

- Re-sending configuration to products when a new version is published; Alertmanager rule (Phase 10)

Phase 8 — Frontend portals

Done

Delivered / scope

- Secure sign-in through Keycloak; OCCPL staff must use their authenticator app before staff features unlock
- Super Admin: dashboard, companies, users and staff roles, products with versioned configuration and features, plans with prices and limits, subscriptions, provisioning jobs with retry, audit log
- Company workspace: company details and settings, contacts, product launcher, subscription, users and invitations with roles, usage against limits, audit log
- Invitation emails lead straight into the portal; works on phones and desktops; keyboard accessible

Verified

- 47 frontend tests (sign-in, permissions, forms, plan editor, user management)
- Browser walkthrough of both portals against the full Docker stack with real sign-in and MFA, including automatic setup of the sample product and a company admin joining from the real email

Open item

- Platform settings page (needs a settings API); names instead of ids in a few lists

Billing (Razorpay)

Done

Delivered / scope

- Invoices are created, numbered and emailed by Razorpay; the company pays on Razorpay's secure page (no card details reach OCCPL systems)
- Renewal invoices go out automatically 7 days before a period ends; staff can also issue one, cancel it, or check its payment status at Razorpay
- A paid invoice renews the subscription for exactly the period it was issued for — never twice; paying during a trial keeps the remaining trial days
- Payment notifications from Razorpay are checked for authenticity and processed once even if repeated; forged notifications are refused and recorded
- Company Billing tab with invoices, Pay link and payments; platform-wide invoice list

Verified

- 7 integration tests against a Razorpay stand-in: one invoice per period, gateway failure, forged and repeated notifications, trial conversion, partial payment, expiry, cancel, status check, automatic renewal invoices
- Browser walkthrough: invoice issued from the portal, payment notification, trial converted to active

Open item

- GST on invoices (decision D-11); one live Razorpay test-mode payment; refunds; separate billing permissions

Phase 9 — Hardening

Done

Delivered / scope

- Automatic check of every API endpoint: each one requires the right sign-in and permission; a company can never reach another company's data (always "not found"), also for new endpoints added later
- Independent security review: 8 issues found and fixed (e.g. a way to exceed the plan's user limit, one sign-in token that every product could reuse, a spoofable client address for rate limiting)
- Known vulnerabilities in libraries: 3 critical and 5 high found and fixed by upgrades; automatic scans in the build pipeline from now on
- Load test with 10,000 companies and 1,000,000 users: all response-time targets met up to 210 requests per second on a small 4-processor test machine; the one slow search was fixed (0.5 s → 0.05 s)
- Backups every night plus every 5 minutes; tested restore of the full data in under a minute, including undoing an accidental deletion to the exact moment before it

Verified

- 5 tenant-isolation checks covering every endpoint; regression tests for every review finding
- Measured results: load-tests/RESULTS.md and infra/backup/DRILL.md

Open item

- Repeat the load test on production-like hardware

Phase 10 — Deployment

Done

Delivered / scope

- Complete AWS setup written as code (Terraform): private network over two data centres, load balancer with HTTPS certificate and web firewall, servers without public access, managed MySQL with automatic standby, managed Redis, email sending for occplsoftech.com, secrets kept in AWS Secrets Manager
- Release pipeline: every change on main is tested and built once, tried on the local Docker setup, and installed on production after an approval, followed by an automatic check; one click to roll back to any earlier release
- Updates replace one server at a time, so production stays available; a failed update returns all servers to the previous release automatically
- Monitoring and alerts by email: errors, slowness, unavailable services, disk, database, failed product set-up, rejected payment notifications, certificate expiry, failed backups
- Backups: database restorable to any second of the last 35 days, plus a daily copy in a second AWS region (Hyderabad); the audit trail can no longer be changed or deleted by anyone
- Runbook: first-time setup, releases and rollback, recovery, incident steps, password rotation

Verified

- Start-up of the production system rehearsed end to end (database set-up, migrations, sign-in with MFA through the public addresses); rollback logic rehearsed; all infrastructure code validated

Open item

- Switch on in AWS (needs account access, DNS zone, SES approval, GitHub settings), then a rollback and a database restore test on production before launch

Risks

#	Risk	Mitigation
R-1	Flyway is officially verified only up to MySQL 9.4	Production uses MySQL 8.4 LTS (verified); 9.7 stays in the test matrix for the later upgrade
R-2	Spring Boot 4 / Jackson 3 ecosystem is new	Prefer Boot-managed libraries; check compatibility before adding any
R-3	Self-hosted Keycloak must be operated	One Keycloak per server, clustered; database backed up with the rest; procedures in the runbook
R-4	Capacity on production hardware unproven	Phase 9 load test met all targets on a small test machine; repeat on the production environment before launch
R-5	Razorpay connection tested only against a stand-in	One test-mode payment with real Razorpay keys before going live (DEPLOYMENT.md)

Open decisions

#	Decision	Needed by	Default if not answered
D-6	Confirm availability and latency targets and expected peak load	Before launch	99.9% availability; p95 under 300 ms for reads (met in the Phase 9 test)
D-7	Can Company Admins buy or upgrade plans themselves?	Phase 6 / 8	Super Admin assigns plans
D-8	Self-registration or invitation only?	Phase 4	Invitation only
D-9	How users open products (subdomains vs one gateway)	Phase 7	Applied: each product has its own address plus a launch path; single sign-on
D-10	Personal-data retention and export obligations	Before launch	Export and deletion workflows, configurable retention
D-11	GST on invoices: are plan prices tax-inclusive, which rate and code, print the company's GSTIN?	Live payments	None applied today: the plan price is the invoice amount